Guided by Parag Sir

Assignment-2

Cyber Security – Introduction to Cybersecurity

Module 1 CS- Introduction

1 what is meaning of cyber security

ANS: - Cyber security means protecting computers, networks, systems, and data from unauthorized access, attacks, damage, or theft. It is the practice of keeping digital information safe from hackers and cyber threats.

2.What are the main objectives of cyber security?

ANS: - the main goals are based on the CIA triad:

- Confidentiality: - ensure data is accessible only to authorized users. Eg: password protection, encryption.
- Integrity: - Ensure data is accurate and not modified illegally. Eg: Hashing, digital signatures.

- Availability: - Ensure systems and data are available when needed. Eg: Backup systems, servers uptime.
  - Other Objectives: - Authentication (verify uptimes), Authorization (access control), Non-repudiation (cannot deny actions).

3.What is offensive and defensive in cyber security?

ANS: - Offensive Security (Red Team): - Attack systems to find vulnerabilities, Done by ethical hackers, activities like Penetration testing (VAPT), Exploitation, Social engineering.

- Defensive Security (Blue Team): - protect systems from attacks, activities like Firewall, Monitoring logs, Incidents response, Antivirus.

4.what is cyberspace and low

ANS: - Cyberspace: - it is the virtual world of the internet and digital communication where users interact. Eg: - websites, Emails, social media, Cloud systems.

- Cyber law: - cyber law refers to legal rules and regulations for internet usage. It includes cybercrime (hacking, fraud), data protection, privacy laws, digital signatures.

5. What is cyber welfare?

ANS: - Cyber Welfare means ensuring safety, rights and well-being of users in the digital world.it includes protecting users from cyberbullying, ensuring privacy, safe internet usage, Awareness about cyber threats.

6.Explain the Types of Hackers

- **Ans: -** White hat hacker (Ethical Hacker): - work legally, help organizations improve security.
- Black hat Hacker: - illegal hackers, steal data, damage systems.
- Grey hat hackers: - Between white and black, works as they wish.
- Script kiddies: - Beginners using ready-made tools, limited knowledge.
- Hacktivists: - Hack for political and social causes.

7.What is the full form of SOC in cyber security

ANS: - SOC = Security Operations Center, it is centralized unit that monitors security threats, detects attacks, responds to incidents.

8.What are the Challenges of Cyber Security

- **ANS: -** Increasing Cyber Attacks: - Ransomware, phishing, malware.
- Lack of Awareness: - users fall into scams easily.
- Advanced Hackers: - use AI and modern tools.
- Data privacy issues: - Sensitive data leaks.
- IoT Vulnerabilities: - Smart devices are less secure.
- Shortage of skilled professionals: - not enough cybersecurity experts.
- Cloud Security Risks: - Misconfigurations in cloud systems.